

Parameter Tuning

ALPR Pipeline: Comprehensive Parameter Tuning & Method Testing Guide

This guide details all parameters per stage in the ALPR pipeline, explaining their exact context, default values, and downstream impacts. It also provides an exhaustive list of alternative methods and algorithms that can be tested across each stage of the system.

Part 1: Stage-by-Stage Tunable Parameters

Stage 1: Frame Ingestion & Decoding

Parameter	Location / Default	Tuning Range / Options	Context & Impact on Pipeline
target_resolution	(1280, 720) or None	(1920, 1080), (1280, 720), None (native)	Controls input frame scaling. Lower resolution speeds up decoding and detection but degrades small license plate character details. None preserves raw sensor resolution.
queue_size	128 frames	32 – 512	Frame buffer size between ingestion thread and detector. Prevents I/O stalls on high-bitrate video streams.
batch_size	1 (or 4 – 8 for batching)	1 – 16	Number of decoded frames passed downstream at once. Higher values enable batched GPU inference in Stage 2 & 2.5.

Stage 2: Vehicle Detection & Multi-Object Tracking (MOT)

Parameter	Default	Tuning Range	Context & Impact on Pipeline
VEHICLE_CLASS_IDS	{2} (Cars only)	{2}, {2, 5, 7} (Cars, Buses, Trucks), {2, 3, 5, 7}	Filter mask for COCO classes. Restricting to {2} eliminates motorcycles and background clutter; expanding enables commercial transport tracking.
DETECTION_CONF_THRESHOLD	0.25	0.15 – 0.60	Vehicle YOLO confidence cutoff. Lowering captures distant/occluded cars but introduces false positives. Raising keeps only high-confidence vehicles.
DETECTION_IOU_THRESHOLD	0.45	0.30 – 0.70	Non-Maximum Suppression (NMS) IoU threshold for vehicle bounding boxes. Prevents duplicate boxes on the same vehicle.
MAX_FRAME_SIZE	1920	1080 – 2560	Resize cap before vehicle detection to conserve GPU VRAM.
TRACK_BUFFER	90 frames (~3s)	30 – 150 frames	ByteTrack lost track persistence. Retains vehicle track IDs during temporary occlusions (trees, signboards, passing vehicles).
TRACK_MATCH_THRESHOLD	0.45	0.20 – 0.80	Minimum association threshold for ByteTrack. Lowering tolerates fast vehicle motion/camera panning; raising prevents ID switching between adjacent vehicles.

Stage 2.5: Full-Frame License Plate Detection

Parameter	Default	Tuning Range	Context & Impact on Pipeline
PLATE_CONF_THRESHOLD	0.30	0.15 – 0.65	YOLO plate detector confidence cutoff. Crucial knob: lowering captures distant/dark plates; raising rejects bumper stickers, grilles, and reflections.
PLATE_DETECT_IMGSZ	1280	640 , 1024 , 1280 , 1920	Input image size for the plate YOLO model. Higher resolutions dramatically improve small plate recall at the cost of GPU inference latency.
MIN_PLATE_AREA	80 px ²	50 – 500 px ²	Minimum pixel area for a valid plate detection. Filters out micro-detections and distant noise.
MIN_PLATE_ASPECT_RATIO	1.10	1.0 – 2.0	Lower aspect ratio limit (width/height). Set to 1.10 to allow 2-line square Indian plates (~1.1–1.4) while rejecting vertical adverts (< 1.0).
MAX_PLATE_ASPECT_RATIO	5.5	3.5 – 6.5	Upper aspect ratio limit. Filters out ultra-wide vehicle chrome trim strips and bumper bars.

Stage 3: Active Vehicle Buffering & Spatiotemporal Association

Parameter	Default	Tuning Range	Context & Impact on Pipeline
PLATE_VEHICLE_IOA_THRESHOLD	0.35	0.20 – 0.60	Intersection-over-Area ($\text{Area}(\text{Plate} \cap \text{Vehicle}) / \text{Area}(\text{Plate})$). Associates a detected plate to a vehicle track. Must not use strict containment or IoU because plates sit entirely inside larger vehicle boxes.

Parameter	Default	Tuning Range	Context & Impact on Pipeline
VEHICLE_BBOX_ASSOC_EXPAND	0.12 (12%)	0.05 – 0.25	Percentage by which the vehicle bounding box is expanded during association. Prevents pla located at the bottom bumper edge from falling outside tight YOLO boxes.
BUFFER_MAX_SIZE	None (unlimited)	20 – 200 or None	Max frames buffered per vehicle. None retains the entire passage of a vehicle for maximum temporal diversity.
BUFFER_TIMEOUT_FRAMES	90 frames	30 – 150 frames	Number of consecutive frames without a vehicle detection before triggering finalization (scene exit).
ROI_PAD	80 px	20 – 150 px	Context padding stored around the vehicle bounding box for high-resolution pipeline cropping.

Stage 4: Vehicle Track Finalization & Sanity Gating

Parameter	Default	Tuning Range	Context & Impact on Pipeline
MIN_FRAMES_FOR_RELIABLE	3 frames	1 – 10 frames	Drops ephemeral 1–2 frame tracking glitches that lack enough history for temporal voting.
AREA_STD_THRESHOLD	5000.0	2000.0 – 10000.0	Standard deviation threshold of vehicle bounding box area. Large fluctuations flag potential ID switches (e.g., track jumps from a distant car to a nearby truck).
DIVERSITY_RATIO_LIMIT	0.95	0.70 – 0.99	Image hash similarity limit. Flags static/parked vehicles with no visual variance across frames.

Stage 5: Job Creation & Frame Quality Selection

Parameter	Default	Tuning Range	Context & Impact on Pipeline
TOP_K_FRAMES	20	3 – 30	Maximum candidate frames selected per vehicle to send to the OCR worker pool. Balances OCR computational cost with temporal sampling density.
BLUR_THRESHOLD	80.0	30.0 – 200.0	Laplacian variance threshold. Vehicle frames below this are flagged as blurry/occluded and skipped in favor of sharper frames.
Scoring Weight Formula	$\sqrt{\text{Area}} \times \text{Sharpness} \times \text{Conf}$	Custom heuristic	Ranks frames having a matched plate by plate size, image sharpness, and detector confidence to prioritize the cleanest frames for OCR.

Stage 6: Asynchronous Worker Pool, Enhancement & OCR

Parameter	Default	Tuning Range	Context & Impact on Pipeline
NUM_WORKERS	3 threads	1 – 8	Number of parallel OCR worker threads. Scales with CPU/GPU capacity.
PLATE_PADDING	10 px	0 – 25 px	Pixel padding added around the plate crop. Ensures outer characters (e.g., initial state codes

Parameter	Default	Tuning Range	Context & Impact on Pipeline
			KA, MH or final digits) are not clipped.
PLATE_UPSCALE_FACTOR	2.0	1.5 – 4.0	Super-resolution scaling factor using Lanczos4 interpolation before OCR. Essential for small plates (< 150px width).
min_width / min_height	120 / 30 px	100-240 / 25-60 px	Floor dimensions for upscaled plate crops fed to EasyOCR.
Bilateral Filter (d, sigmaColor, sigmaSpace)	(11, 17, 17)	d: 5-15, \sigma: 10-50	Bilateral smoothing parameters. Removes background plate noise while preserving crisp character edges.
Top-Hat/Black-Hat Kernel	(9, 5) rectangular	(5, 3) to (15, 7)	Morphological structuring element in Adaptive Branch. Isolates dark embossed characters from bright reflective plate backgrounds.
CLAHE clipLimit & tileGridSize	2.5, (6, 6)	1.5-4.0, (4,4)-(8,8)	Contrast Limited Adaptive Histogram Equalization settings for localized contrast enhancement.
plate_edge_density Threshold	0.02 (2%)	0.01 – 0.05	Canny edge pixel density. Discards blank bumpers or smooth plastic that lack text edges before running OCR.

Stage 7: Multi-Frame Temporal Fusion & Post-Processing

Parameter	Default	Tuning Range	Context & Impact on Pipeline
MIN_READINGS_FOR_FUSION	2	1 – 5	Minimum number of valid per-

Parameter	Default	Tuning Range	Context & Impact on Pipeline
			frame OCR readings required to trigger multi-frame character voting. If fewer, falls back to the highest-confidence single reading.
Positional Priors & Soft Snapping	Hardcoded in <code>plate_utils.py</code>	State Aliases / Levenshtein ≤ 1	State code fuzzy matching, RTO digit forcing, middle series letter forcing, and registration digit forcing.

Stage 9: Trajectory & Bounding-Box Interpolation

Parameter	Default	Tuning Range	Context & Impact on Pipeline
Interpolation Kind	<code>linear</code>	<code>linear</code> , <code>quadratic</code> , <code>cubic</code> , <code>nearest</code>	Mathematical method used to reconstruct missing vehicle and plate coordinates across intermediate frames.
Max Gap Limit	Unlimited	<code>5</code> - <code>30</code> frames	Maximum missing frame gap across which interpolation is permitted (prevents bridging disconnected tracks).

Part 2: Method Testing Matrix & Alternative Algorithms

Beyond parameter values, different algorithms and processing methods can be tested at specific stages across the pipeline:

METHOD TESTING PLAYGROUND		
Stage	Currently Implemented Method	Alternative Methods to Test
1. Detection	ByteTrack	BoT-SORT, DeepOC-SORT, StrongSORT
2. Tracking	ByteTrack	BoT-SORT, DeepOC-SORT, StrongSORT
2.5 Plate Detection	YOLOv8/v11 (Single Full-Frame)	SAHI (Slicing), RT-DETR, Faster RCNN
3. Association	IoA (Intersection-over-Area)	Generalized IoU, Hungarian Bipartite
5. Frame Selection	Laplacian Variance Sharpness	Tenengrad, FFT High-Freq, BRISQUE
6. Deskewing	MinAreaRect Contours	Hough Line Transform, Spatial Trans
6. Plate Preprocessing	Bilateral Filter & Adaptive Morph	(See Exhaustive Preproc List Below)
6. OCR Engine	EasyOCR (PyTorch CRNN)	PaddleOCR, Tesseract, TrOCR, CRNN
7. Temporal Fusion	Conf-Weighted Character Voting	CTC Beam Search, Edit-Dist Medoid
9. Interpolation	1D Linear Interpolation	Kalman Smoother, Akima Splines

Deep Dive: Exhaustive Plate Preprocessing & Enhancement Methods (Stage 6)

The plate enhancement module ([plate_utils.py](#)) is the primary area for testing image-processing techniques to improve OCR accuracy:

1. Color Space & Channel Extraction

- **Grayscale Conversion:** Standard luminosity formula ($0.299R + 0.587G + 0.114B$).
- **L*a*b* Color Space (L-Channel):** Isolates lightness from chromatic variations, mitigating color cast from headlights/streetlights.
- **YCrCb (Y-Channel):** Isolates luminance for lighting correction.
- **HSV/HSB (V/L Channel with Saturation Masking):** Useful for distinguishing yellow commercial plates from white private plates.

2. Denoising & Edge-Preserving Smoothing

- **Bilateral Filtering (Current Branch A):** $\mathcal{O}(N)$ spatial + radiometric smoothing; preserves sharp character boundaries.
- **Non-Local Means (NL-Means) Denoising:** Computes patch similarity across the crop; removes sensor noise in low-light/high-ISO frames.
- **Guided Filter:** Edge-preserving smoothing guided by the original high-contrast image; avoids gradient reversal artifacts.
- **Gaussian Blur vs. Median Blur:** Median blur removes salt-and-pepper noise and small bolt-head artifacts.

3. Contrast, Illumination & Morphological Equalization

- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Prevents over-amplification of noise in homogeneous plate regions.
- **Top-Hat & Black-Hat Morphological Filtering (Current Branch B):**

$$\text{Enhanced} = \text{Image} + \text{TopHat}(\text{Image}) - \text{BlackHat}(\text{Image})$$

Isolates dark embossed characters from bright reflective backgrounds regardless of gradient shadows.

- **Gamma Correction (Power Law):** Dynamic gamma adjustments ($\gamma \in [0.4, 2.5]$) based on mean image brightness.
- **Retinex Algorithm (Single-Scale or Multi-Scale MSRCR):** Separates illumination reflectance components to eliminate harsh shadow lines across the plate.

4. Binarization & Thresholding Techniques

- **Otsu's Global Binarization:** Minimizes intra-class variance of foreground/background pixels.
- **Adaptive Gaussian / Mean Thresholding:** Computes localized thresholds for varying illumination across the plate.
- **Sauvola & Niblack Thresholding:** Specialized document/text binarization based on local standard deviation (handles stained/degraded plates).
- **Wolf-Jolion Binarization:** Variant of Sauvola addressing low-contrast background degradation.

5. Super-Resolution & Rescaling

- **Lanczos4 Interpolation (Current Default):** 8×8 sinc interpolation; minimizes aliasing on character strokes.
- **Bicubic vs. Area-relation Interpolation:** Faster alternatives for moderate scale factors.
- **Deep Learning Super-Resolution (Real-ESRGAN / EDSR / FSRCNN):** Neural upscaling for low-resolution plates ($< 80\text{px}$ width).

6. Geometric Deskewing & Perspective Rectification

- **Minimum Area Bounding Box Rotation (Current Implementation):** Affine rotation based on contour orientation.
- **Hough Line Transform Deskewing:** Detects upper and lower plate borders to compute precise tilt angles.

- **4-Point Perspective Transform (Homography):** Warps non-rectangular quadrilateral plates to a standardized rectangular ratio (e.g., $500 \times 120\text{px}$).

Alternative OCR Engines & Architectures (Stage 6)

OCR Engine	Architecture	Strengths	Weaknesses
EasyOCR <i>(Current)</i>	CRAFT (Text Detection) + ResNet + BiLSTM + CTC (Recognition)	Strong out-of-the-box accuracy on alphanumeric characters; GPU accelerated.	Slower inference CPU; occasional space/symbol hallucination.
PaddleOCR (PP-OCRv4)	DBNet + MobileNetV3 + SVTR / SAR	Ultra-fast inference (~5-15ms per crop); highly optimized for multilingual and vehicle plates.	Requires <code>paddlepaddle</code> dependency.
Tesseract 5 (LSTM)	Line finding + BiLSTM text line recognizer	Fast on CPU; highly configurable character whitelists (<code>tessedit_char_whitelist</code>).	Sensitive to crop border noise, rotation, and non uniform lighting.
TrOCR (Transformer OCR)	Vision Transformer (ViT) Encoder + RoBERTa Decoder	High accuracy on degraded/distorted text; captures complex character contexts.	Computationally intensive; higher latency per crop
Custom CRNN / LPRNet	Lightweight CNN + CTC trained on Indian number plates	Real-time throughput (> 200 FPS); zero external character hallucinations.	Requires custom synthetic/annotated Indian plate training dataset.

Alternative Temporal Fusion & Error Correction Methods (Stage 7)

1. **Confidence-Weighted Character Majority Voting** *(Current)*:

- Aligns readings by modal length and votes on character slots using OCR confidence weights combined with positional priors.
- Levenshtein Distance Medoid Consensus:**
 - Computes the pairwise normalized Levenshtein distance matrix across all candidate strings and selects the string that minimizes total edit distance.
 - CTC Softmax Logit Alignment & Merging:**
 - Directly averages the raw output probability distributions across frames at each character position before CTC decoding.
 - Finite State Transducer (FST) / RegEx Beam Search:**
 - Decodes characters using a beam search constrained by Indian HSRP grammar rules (e.g., State Code → District Code → Series → 4 Digits).

Part 3: Recommended Parameter Tuning Matrix

To systematically optimize pipeline accuracy, tune parameters in this recommended sequence:

Priority	Stage	Parameter / Method	Recommended Tuning Sweep	Optimization Metric
P1	Stage 2.5	PLATE_CONF_THRESHOLD & PLATE_DETECT_IMGSZ	Conf: [0.20, 0.30, 0.40] Imgsz: [1024, 1280, 1920]	Plate Recognition & Localization Precision
P2	Stage 6	Preprocessing: Bilateral vs. Adaptive CLAHE vs. Sauvola	Branch comparison across test video	Per-frame OCR Accuracy, Character Error Rate (CER)
P3	Stage 6	PLATE_PADDING & PLATE_UPSCALE_FACTOR	Pad: [5, 10, 15, 20] px Upscale: [1.5, 2.0, 3.0]	Edge Character Accuracy (State & final)

Priority	Stage	Parameter / Method	Recommended Tuning Sweep	Optimization Metric
P4	Stage 3	PLATE_VEHICLE_IOA_THRESHOLD & VEHICLE_BBOX_ASSOC_EXPAND	IoA: [0.25, 0.35, 0.45] Expand: [0.08, 0.12, 0.18]	Plate-to-Vehicle Association Rate
P5	Stage 5	TOP_K_FRAMES & BLUR_THRESHOLD	K: [10, 15, 20, 25] Blur: [50.0, 80.0, 120.0]	Pipeline Process FPS vs. Fusion Quality
P6	Stage 7	MIN_READINGS_FOR_FUSION & State Code Fuzzy Snapping	Readings: [2, 3, 4]	Final Fusion Accuracy Ground Truth Match Rate